

SE LAB 4

Prompt:

as a developer and a user myself i have noticed a few features that can improve the game rules and its usability, here are the following changes that should be made to improve the game:

Task 1 — Consistent coordinate handling

Make coordinate representation consistent across input, ship placement, player shots, and AI shots so occupied cells are recognised correctly.

Done when: every real ship coordinate is treated as a hit and every known empty coordinate is treated as a miss.

Task 2 — Complete fleet and win logic

Support multiple ships, track individual ship damage, identify sunk ships, prevent repeated shots, and end the game when the fleet is sunk.

Task 3 — Improve the AI

After a hit, make the AI preferentially consider nearby untried cells. It must never shoot the same coordinate twice and must handle a board with no remaining choices.

Task 4 — Shot-level feedback

Ensure hit/miss/sunk feedback occurs exactly once for an actual shot. Internal AI candidate selection must not produce false hit/miss messages.

also for the testing team provide appropriate test cases in a separate text file such as Test hits, misses, repeated shots, sinking one ship, sinking all ships, AI repeated-shot prevention, adjacent targeting after a hit, invalid coordinates, and quitting.

also try adding a display showing all the possible options to fire at and use a X to show hits, O to show misses and _ to show available to hit

do this one task at a time and commit to the repo https://github.com/shantanu666/19_battleship.git and ask me before proceeding to the next task

